

VOLTAGE AND CURRENT DIVIDER THEOREM

CIRCUIT ANALYSIS METHOD

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TOPIC OUTLINE

Voltage Divider Theorem (VDT)

Current Divider Theorem (IDT)



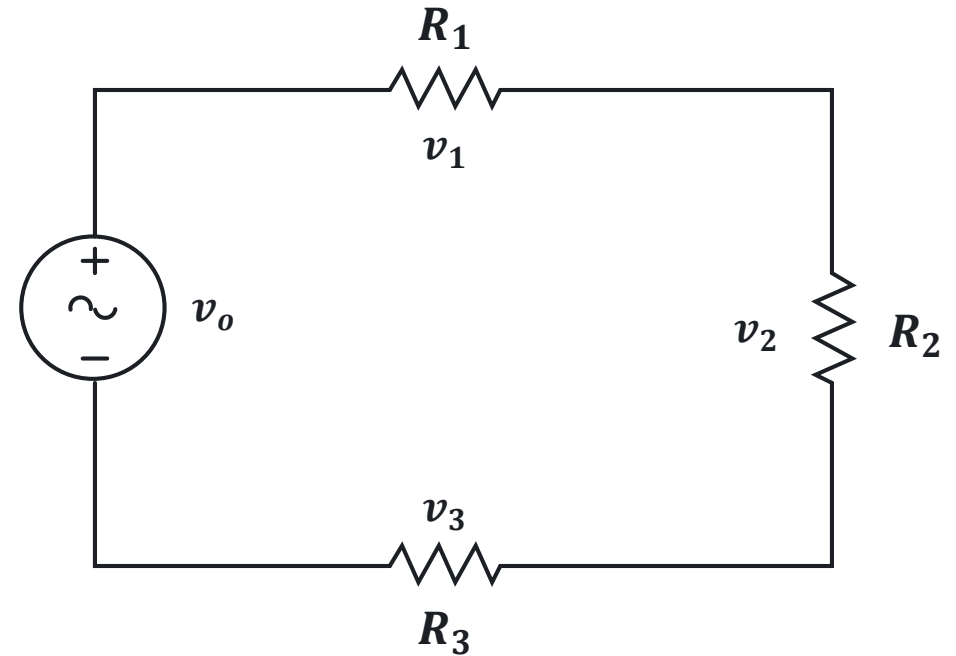
VOLTAGE DIVIDER THEOREM



VOLTAGE DIVIDER THEOREM

A voltage divider is consists of two or more resistors connected in series with a voltage source. The voltage across each resistor is a fraction of the total applied voltage, determined by the ratio of that resistor's value to the total resistance in the series.

Series Network



VOLTAGE DIVIDER THEOREM

Series Network

$$v_1 = v_o \frac{R_1}{R_o}$$

$$v_2 = v_o \frac{R_2}{R_o}$$

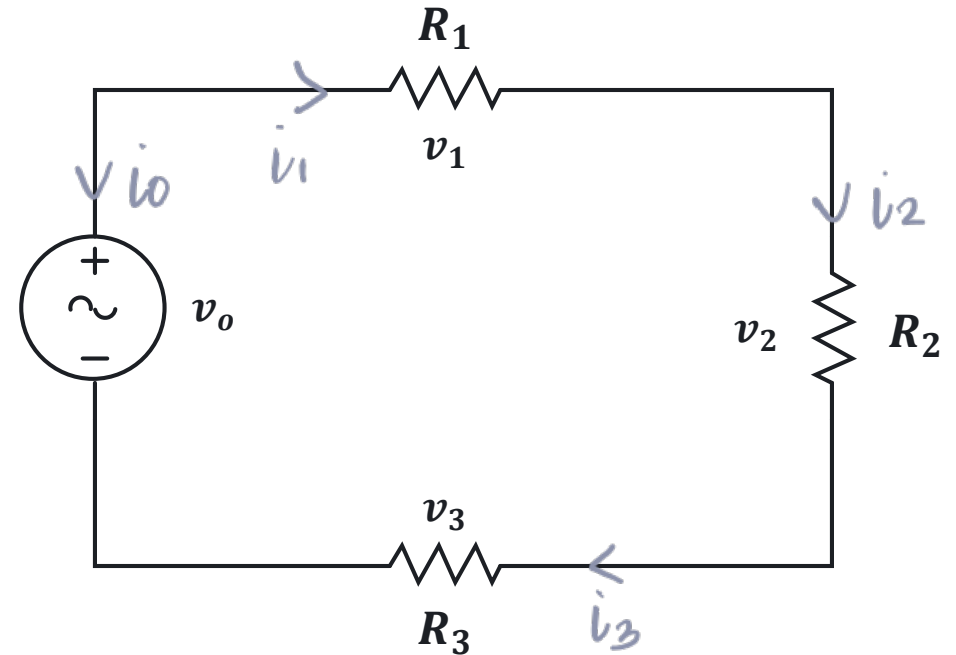
$$v_3 = v_o \frac{R_3}{R_o}$$

$$v_n = v_o \frac{R_n}{R_o}$$

$$i_o = i_1 = i_2 = i_3$$

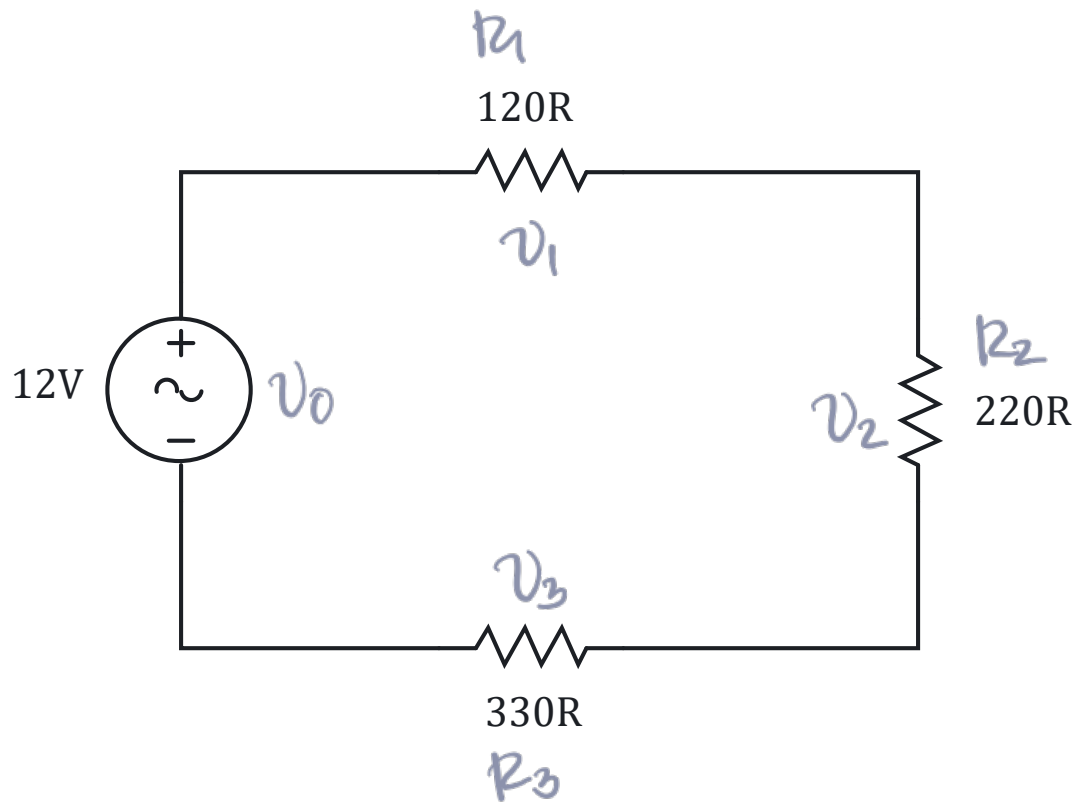
$$\frac{v_o}{R_o} = \frac{v_1}{R_1}$$

$$v_1 = \frac{v_o R_1}{R_o}$$



EXERCISE

Determine the **voltage** drop across each resistor of the given circuit.



Solution

Total resistance

$$R_0 = R_1 + R_2 + R_3$$

$$R_0 = 120 + 220 + 330$$

$$R_0 = 670\Omega$$

$$V_1 = V_0 \frac{R_1}{R_0}$$

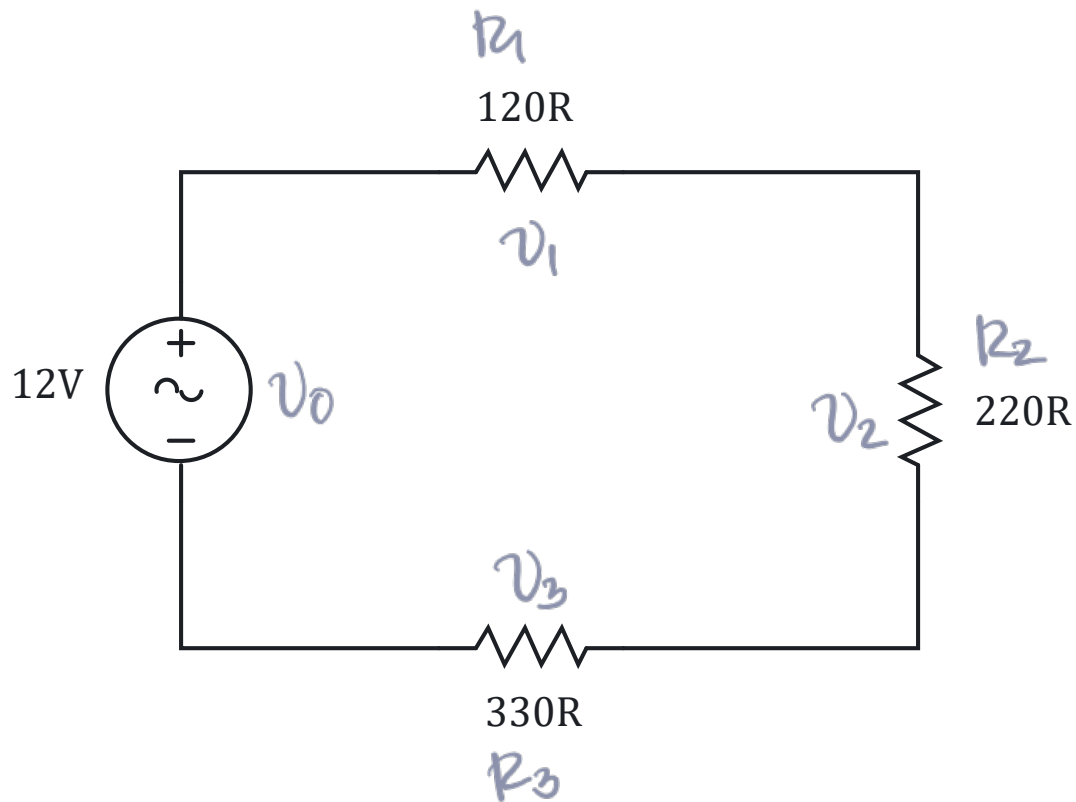
$$V_1 = 12 \frac{120}{670}$$

$$V_1 = 2.15V$$

ans

EXERCISE

Determine the **voltage** drop across each resistor of the given circuit.



Solution

$$V_2 = V_0 \frac{R_2}{R_0}$$

$$V_2 = 12 \frac{220}{670}$$

$$V_2 = 3.94V$$

ans

$$V_3 = V_0 \frac{R_3}{R_0}$$

$$V_3 = 12 \frac{330}{670}$$

$$V_3 = 5.91V$$

ans



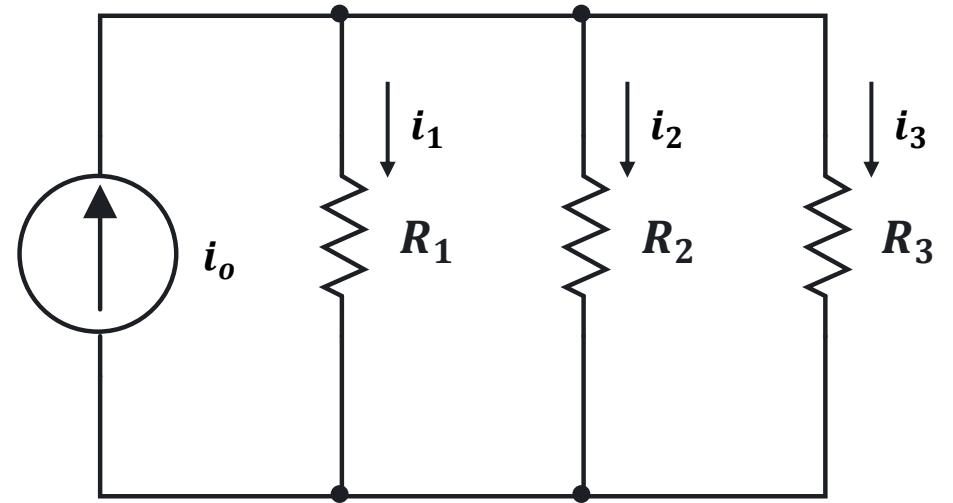
CURRENT DIVIDER THEOREM



CURRENT DIVIDER THEOREM

A current divider is consists of two or more resistors connected in parallel with a current source. The current through each resistor is a fraction of the total current, determined by the ratio of the other resistor's value to the total resistance in the parallel combination.

Parallel Network



CURRENT DIVIDER THEOREM

Parallel Network

$$i_1 = i_o \frac{R_o}{R_1}$$

$$i_2 = i_o \frac{R_o}{R_2}$$

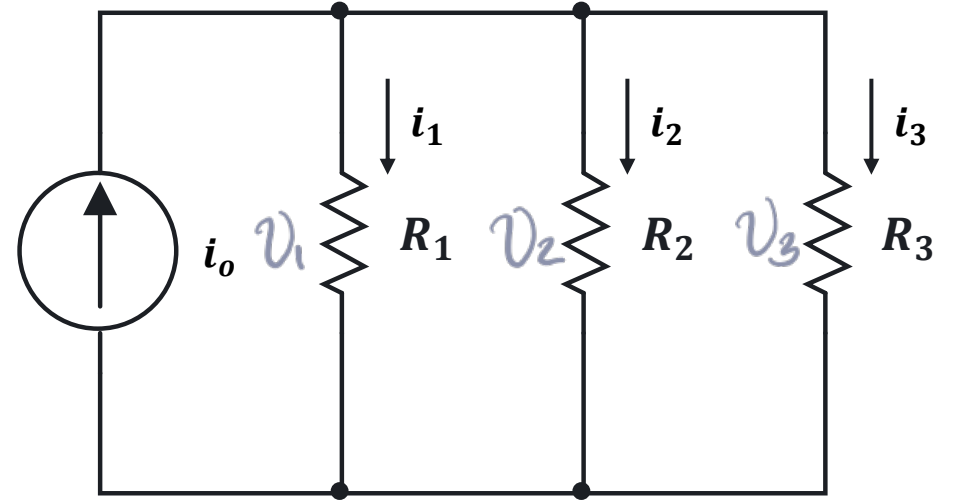
$$i_3 = i_o \frac{R_o}{R_3}$$

$$i_n = i_o \frac{R_o}{R_n}$$

$$v_o = v_1 = v_2 = v_3$$

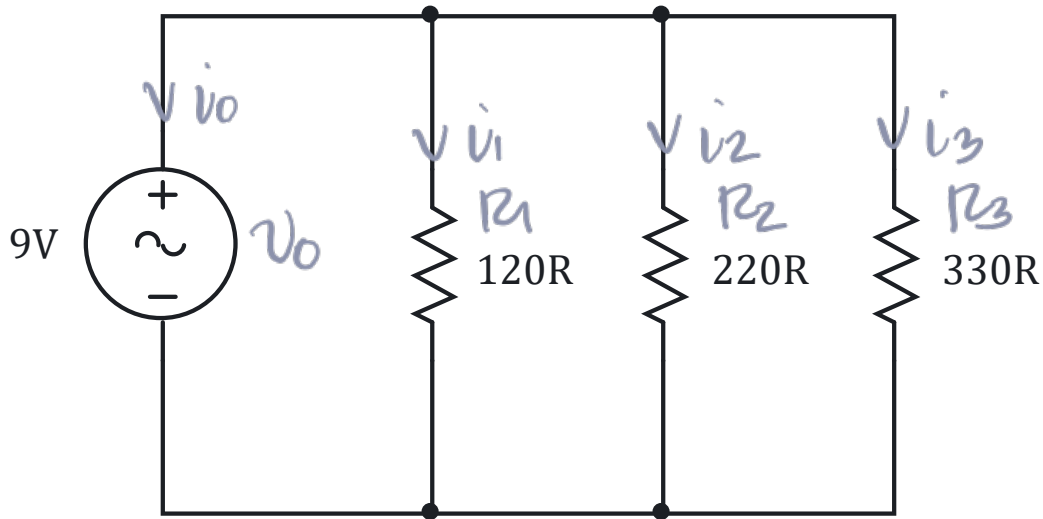
$$\frac{i_o R_o}{R_1} = \frac{i_1 R_1}{R_1}$$

$$i_1 = \frac{i_o R_o}{R_1}$$



EXERCISE

Determine the **current** flowing through each resistor of the given circuit.



Solution

Total Resistance

$$R_0 = R_1 + R_2 + R_3$$

$$\frac{1}{R_0} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_0} = \frac{1}{120} + \frac{1}{220} + \frac{1}{330}$$

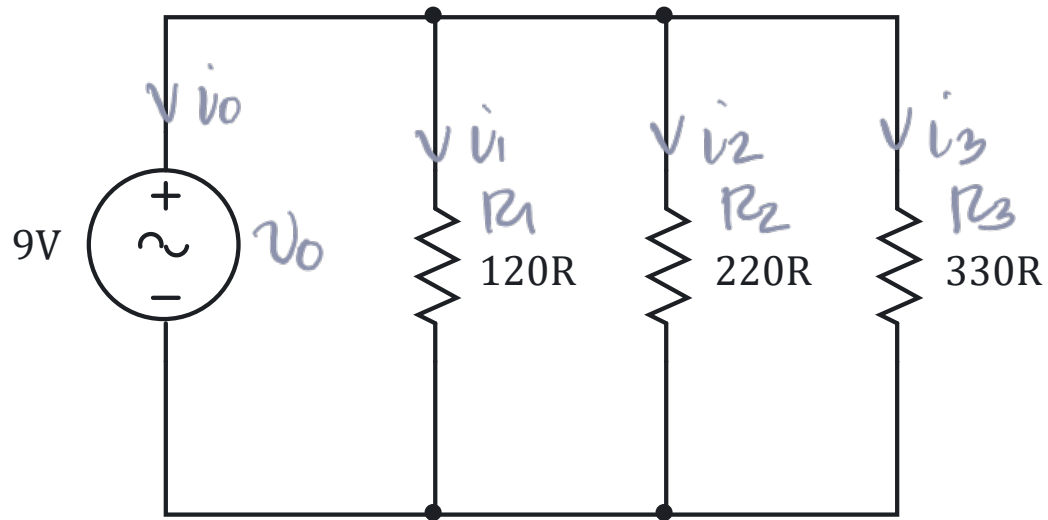
$$\frac{1}{R_0} = \frac{7}{440} \quad R_0 = \frac{440}{7}$$

$$\underline{R_0 = 62.86 \, \Omega}$$



EXERCISE

Determine the **current** flowing through each resistor of the given circuit.



Solution

Total Current

$$i_0 = \frac{v_0}{R_0}$$

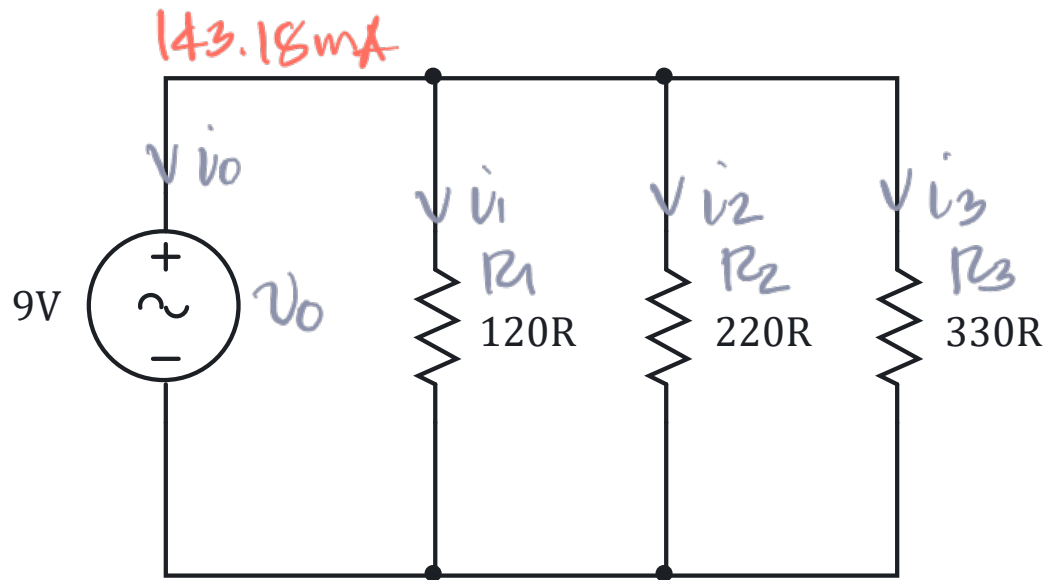
$$i_0 = \frac{9}{62.86}$$

$$i_0 = 143.18 \text{ mA}$$



EXERCISE

Determine the **current** flowing through each resistor of the given circuit.



Solution

$$i_1 = i_0 \frac{R_0}{R_1}$$

$$i_1 = 143.18 \text{ m} \frac{62.86}{120}$$

$$i_1 = 75 \text{ mA}$$

ans

$$i_2 = i_0 \frac{R_0}{R_2}$$

$$i_2 = 143.18 \text{ m} \frac{62.86}{220}$$

$$i_2 = 40.91 \text{ mA}$$

ans

$$i_3 = i_0 \frac{R_0}{R_3}$$

$$i_3 = 143.18 \text{ m} \frac{62.86}{330}$$

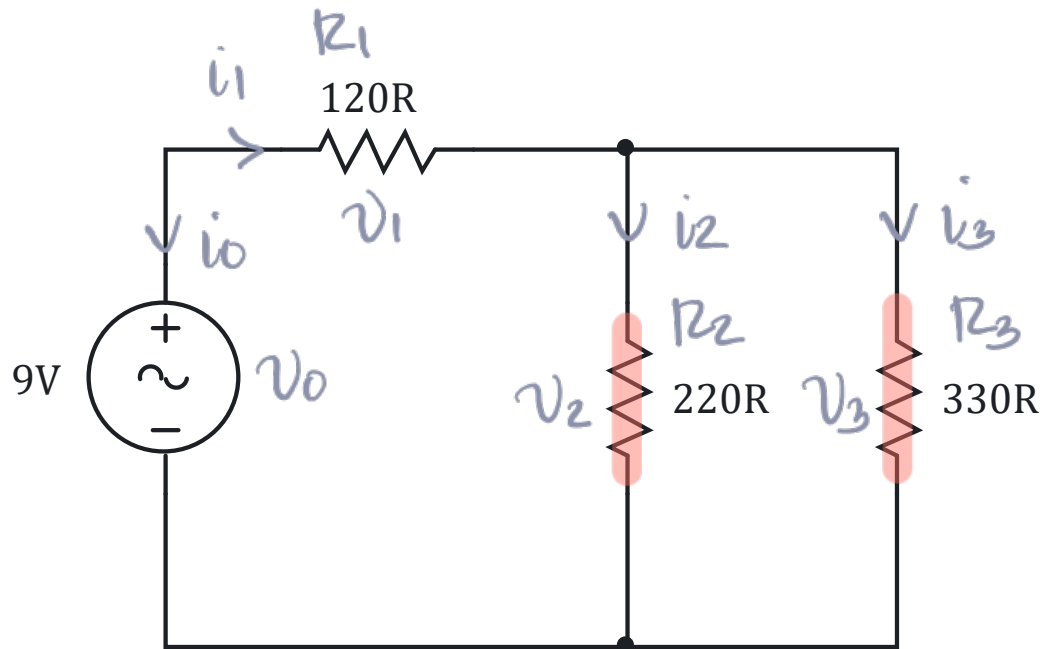
$$i_3 = 27.27 \text{ mA}$$

ans



EXERCISE

Determine the **current** flowing through each resistor and the **voltage** drop across each resistor of the given circuit.



Solution

Equivalent Resistance of R_2 and R_3

$$G_{2,3} = G_2 + G_3$$

$$\frac{1}{R_{2,3}} = \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_{2,3}} = \frac{1}{220} + \frac{1}{330}$$

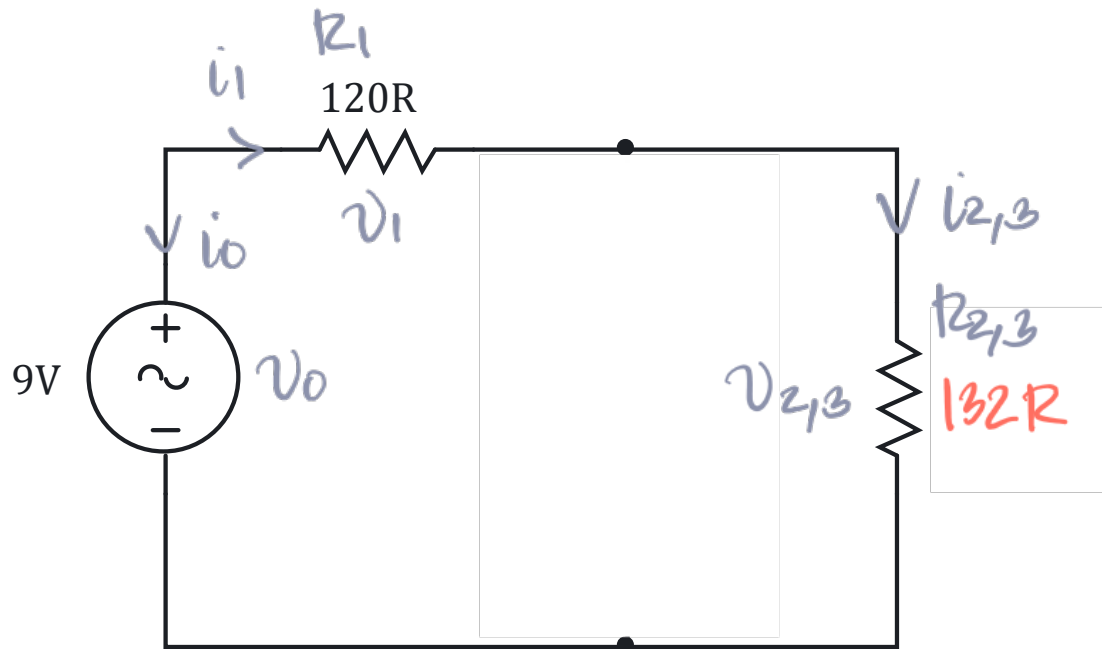
$$\frac{1}{R_{2,3}} = \frac{1}{132}$$

$$\underline{R_{2,3} = 132\Omega}$$



EXERCISE

Determine the **current** flowing through each resistor and the **voltage** drop across each resistor of the given circuit.



Solution

$$R_0 = R_1 + R_{2,3}$$

$$R_0 = 120 + 132$$

$$R_0 = 252\Omega$$

by VOT

$$V_1 = V_0 \frac{R_1}{R_0}$$

$$V_1 = 9 \frac{120}{252}$$

$$V_1 = 4.29\text{ V}$$

ans

$$i_1 = \frac{V_1}{R_1}$$

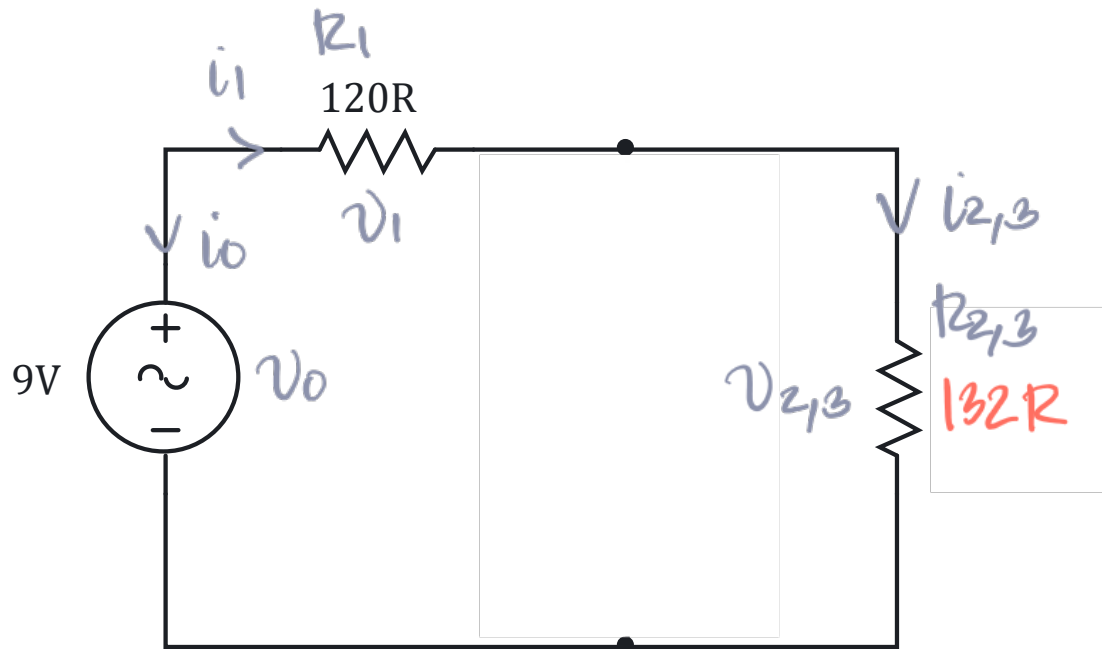
$$i_1 = \frac{4.29}{120}$$

$$i_1 = 35.75\text{ mA}$$

ans

EXERCISE

Determine the **current** flowing through each resistor and the **voltage** drop across each resistor of the given circuit.



Solution

$$V_{2,3} = V_0 \frac{R_{2,3}}{R_0}$$

$$V_{2,3} = 9 \frac{132}{252}$$

$$\underline{V_{2,3} = 4.71 \text{ V}}$$

$$\boxed{V_2 = 4.71 \text{ V}}$$

ans

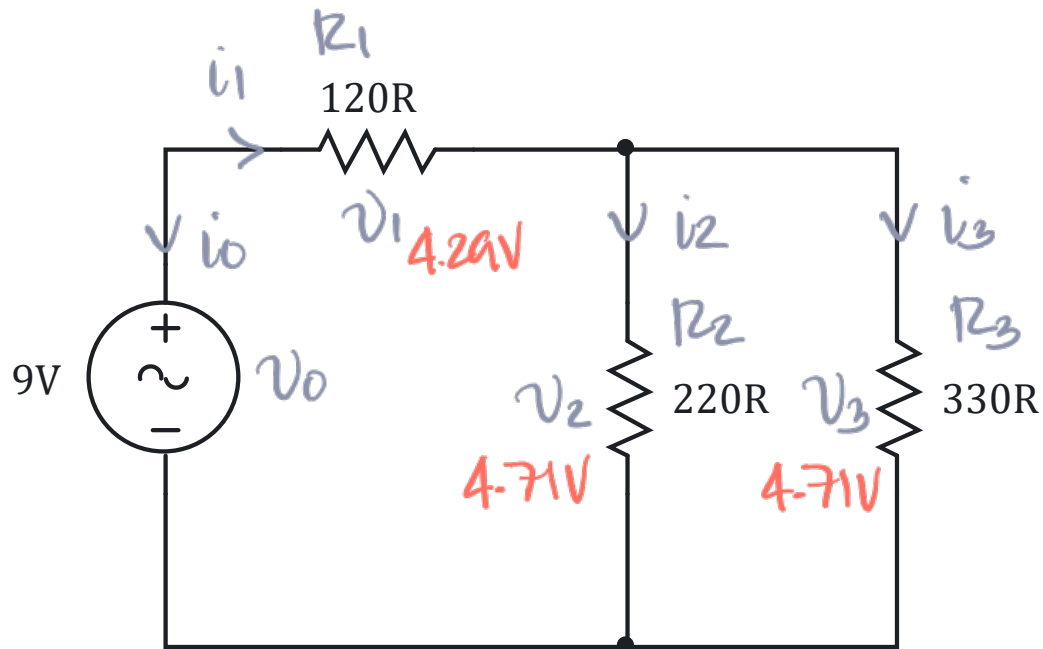
$$\boxed{V_3 = 4.71 \text{ V}}$$

ans



EXERCISE

Determine the **current** flowing through each resistor and the **voltage** drop across each resistor of the given circuit.



Solution

$$i_2 = \frac{V_2}{R_2}$$

$$i_2 = \frac{4.71}{220}$$

$$i_2 = 21.41 \text{ mA}$$

ans

$$i_3 = \frac{V_3}{R_3}$$

$$i_3 = \frac{4.71}{330}$$

$$i_3 = 14.27 \text{ mA}$$

ans



LABORATORY

